MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION, OCTOBER 2025

(2024 ADMISSION ONWARDS)

MG3DSCMBG203 – FoodborneDiseases, Food Sanitation and Food Safety

Duration: 1.5Hrs Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** mark

1. Which of the following is NOT a physical hazard in the food industry? [1] [U]
   a) Bacteria b) Metal pieces c) Plastic pieces d) Hair
2. Which of the following is a naturally occurring chemical hazard in food? [1] [U]
   a) Monosodium glutamate b) Emulsifiers c) Aflatoxins d) Antioxidants
3. Heavy metals like lead and mercury in food are examples of: [1] [U]
   a) Biological hazards b) Physical hazards c) Chemical hazards d) Nutritional additives
4. Which bacterium causes foodborne intoxication through pre-formed neurotoxins, often linked to improperly canned foods? [2] [U]
   a) *Staphylococcus aureus*b) *Clostridium botulinum* c) *Bacillus cereus* d) *Escherichia coli*
5. Which protozoan parasite is responsible for amoebic dysentery from contaminated food or water? [2][U]
   a) *Giardia lamblia* b) *Entamoebahistolytica*

c) *Cryptosporidium parvum* d) *Cyclosporacayetanensis*

6. What is considered a physical sanitiser in food processing? [3][U]

a) Quaternary ammonium compounds b) Steam at high temperatures

c) Peroxyacetic acid d) Chlorine-based sanitiser

7. Identify which of these is not typically part of QC responsibilities? [4] [A]

a) Microbiological testing of products

b) Product inspection and defect detection

c) Auditing production processes to prevent future issues

d) Chemical and physical analyses of raw and final products

8. Identify the activated sludge process, commonly used to treat wastewater in food processing plants. [4] [A]

a) Anaerobic sludge blanket b) Biological aerobic suspended-growth

c) Trickling filter attached-growth, d) Chemical coagulation process

9. What does HACCP stand for? [4] [U]

a) Hazard Analysis and Control of Critical Packaging

b) Hazard Analysis Critical Control Point

c) Health Analysis Critical Control Protocol

d) Hazard Analysis of Control Critical Point

10. AGMARK certification is primarily associated with: [4] [U]

a) Packaged foods and fast-food outlets

b) Agricultural produce quality and grading

c) Meat processing standards

d) International ISO certification

(1×10=10)

Part B

Fill-in-the-blank questions

Answer all questions

Each question carries **1** mark

11. The most common cause of acute foodborne viral gastroenteritis worldwide is _____. [1] [U]

12. Cholera is caused by the bacterium _____. [2] [U]

13. CIP stands for Clean_____ , a method of cleaning interior surfaces of pipes, tanks, and equipment without disassembly. [3] [U]

14. QC involves conducting microbiological, chemical, and _____ tests on finished products. [4] [U]

15. GMP stands for Good ____ Practices, covering hygiene, equipment maintenance, personnel training, and sanitation. [4] [U]

(1×5=5)

Part C

True or false questions

Answer all questions

Each question carries **1** mark

16. Norovirus is the leading cause of foodborne illness worldwide. [2] [U]

17. Only bacteria, but not viruses or parasites, cause foodborne illnesses. [2] [U]

18. Apply clinical knowledge to the management of salmonellosis [3] [A]

19. FSSAI is the statutory authority that sets science-based food safety and quality standards in India. [4] [A]
20. Quality Control (QC) teams within food processing units are responsible for verifying that finished products meet regulatory and internal safety standards. [4] [U]

(1×5=5)

Part D

One-word answer
questions.Answer all questions

Each question carries **1** mark

21. Which hazard type includes pesticides, cleaning agents, and natural toxins? [1] [U]

22. What type of toxin produced by *Staphylococcus aureus*causes rapid-onset vomiting? [2] [U]

23. Which virus causes liver inflammation via faecal–oral transmission? [2] [U]

24. What virus is the most common cause of foodborne viral gastroenteritis globally? [2] [U]

25. What is bacillary dysentery? [1] [U]

26. What is an autoclave? [3] [U]

27. Expand COP and CIP. [3] [U]

28. Identify two chemical sanitisers that often target and destroy pathogens. [3] [A]

29. Why is sanitation important in pest control? [3] [U]

30. What is the core function of FSSAI? [3] [U]

31. What is the main function of AGMARK? [3] [U]

32. What is a CCP in the HACCP system? [3] [U]

(1×10=10)

Part E

Short answer questions

Answer any 4 questions

Each question carries 3 marks

33. Explain the three types of hazards found in the food industry [1] [U]

34. What are the symptoms of Botulism? [2] [U]

35. Comment on salmonellosis. [2] [U]

36. Explain the steps for preventing foodborne outbreaks. [2] [U]

37. Examine the role of AGMARK in ensuring the quality of agricultural commodities in India [3] [U]

38. Identify the importance of food safety symbols on packaging. [3] [A]

(3×4=12)

Part F

Long answer questions

Answer any 1 question

Each question carries 8 marks

39. Sketch the mechanisms and management of two major foodborne intoxications: *Staphylococcus aureus*food intoxication and Botulism [1] [A]

40. Explain the seven principles of HACCP and how they support a preventive approach to food safety. [4] [U]

(1×8=8)